Band-1 Hadamard order 13964 – Classical Paley-I (q=13963); IN SPK via Paley-I; Gap 2052 Still OPEN

H_13964 Paley-I PASS; Paley-II FAIL/SKIP (q=6981=3*13*179 NOT prime; ===1 mod 4 but composite); Kronecker FAIL/SKIP (4*3491; 3491===3 mod 4 NOT Hadamard-admissible; 3491 (odd prime)); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-07

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. 2026-10-07_Band1_Hadamard_13932_PaleyI; 2026-10-07_Band1_Hadamard_13908_PaleyI; 2026-10-07_Band1_Hadamard_13904_PaleyI; 2026-10-07_Band1_Hadamard_13884_PaleyI; 2026-10-07_Band1_Hadamard_13880_PaleyI; 2026-10-07_Band1_Hadamard_13860_PaleyI; 2026-10-07_Band1_Hadamard_13832_PaleyI; 2026-10-07_Band1_Hadamard_13808_PaleyI; 2026-10-07_Band1_Hadamard_13800_PaleyI; 2026-10-07_Band1_Hadamard_13764_PaleyI; 2026-10-07_Band1_Hadamard_13760_PaleyI; 2026-10-07_Band1_Hadamard_13752_PaleyI; 2026-10-07_Band1_Hadamard_13724_PaleyI; 2026-10-07_Band1_Hadamard_13712_PaleyI; 2026-10-07_Band1_Hadamard_13692_PaleyI; 2026-10-07_Band1_Hadamard_13688_PaleyI; 2026-10-06_Band1_Hadamard_13680_PaleyI; 2026-10-06_Band1_Hadamard_13628_PaleyI; 2026-10-06_Band1_Hadamard_13620_PaleyI; 2026-10-06_Band1_Hadamard_13592_PaleyI; 2026-10-06_Band1_Hadamard_13568_PaleyI; 2026-10-06_Band1_Hadamard_13524_PaleyI; 2026-10-06_Band1_Hadamard_13500_PaleyI; 2026-10-06_Band1_Hadamard_13488_PaleyI; 2026-10-06_Band1_Hadamard_13464_PaleyI; 2026-10-06_Band1_Hadamard_13452_PaleyI; 2026-10-06_Band1_Hadamard_13412_PaleyI; 2026-10-06_Band1_Hadamard_13400_PaleyI; 2026-10-06_Band1_Hadamard_13368_PaleyI; 2026-10-06_Band1_Hadamard_13340_PaleyI; 2026-10-06_Band1_Hadamard_13332_PaleyI; 2026-10-06_Band1_Hadamard_13328_PaleyI; 2026-10-06_Band1_Hadamard_13292_PaleyI; 2026-10-06_Band1_Hadamard_13268_PaleyI; 2026-10-06_Band1_Hadamard_13260_PaleyI; 2026-10-06_Band1_Hadamard_13220_PaleyI; 2026-10-06_Band1_Hadamard_13188_PaleyI; 2026-10-06_Band1_Hadamard_13184_PaleyI; 2026-10-06_Band1_Hadamard_13172_PaleyI; 2026-10-05_Band1_Hadamard_13164_PaleyI; 2026-10-05_Band1_Hadamard_13160_PaleyI; 2026-10-05_Band1_Hadamard_13152_PaleyI; 2026-10-05_Band1_Hadamard_13148_PaleyI; 2026-10-05_Band1_Hadamard_13128_PaleyI; 2026-10-05_Band1_Hadamard_13104_PaleyI; 2026-10-05_Band1_Hadamard_13100_PaleyI; 2026-10-05_Band1_Hadamard_13064_PaleyI; 2026-10-05_Band1_Hadamard_13044_PaleyI;

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2026-09-23_Band1_Hadamard_10160_PaleyI; 2026-09-23_Band1_Hadamard_10152_PaleyI;
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2026-09-23_Band1_Hadamard_10104_PaleyI; 2026-09-23_Band1_Hadamard_10100_PaleyI;
2026-09-23_Band1_Hadamard_10068_PaleyI.

Lab. `/workspace/experiments/band1/hadamard_13964_paleyI/`.
Cutoff. 2026-10-08T01:18:26Z (~9:18PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 13964 only (verified max|HHˆT-nI|=0), via classical Paley-I q=13963 (prime ===3 mod 4). 13964 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 13964: Paley-I **PASS** ($q = 13963$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=6981=3*13*179 NOT prime; ===1 mod 4 but composite); Kronecker **FAIL/SKIP** (note $13964 = 4 \cdot 3491$; $3491 \equiv 3 \pmod 4$ NOT Hadamard-admissible; 3491 (odd prime)). **13964**

folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **13908+13904+13884+13880+13860+13832+13808+13800+13764+13760+13752+13724+13712+136** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **13964**.

- H_{13964} = Paley-I ($q = 13963$) – classical SPK
- Paley-II for 13964: **FAIL/SKIP** ($q = 6981 = 3 \cdot 13 \cdot 179$ NOT prime; $\equiv 1 \pmod 4$ but composite)
- Kronecker for 13964: **FAIL/SKIP** ($4 \cdot 3491$; $3491 \equiv 3 \pmod 4$ NOT Hadamard-admissible; 3491 (odd prime))
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_13964.py`

Results. overall_pass=True; units 625/625; row blocks 110/110 (block size 128); wall 2530.4571s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-08T01:18:26Z (~9:18PM ET)

Arithmetic

- Target $n = 13964 = q + 1$ with $q = 13963$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 6981 = 3 \cdot 13 \cdot 179$ NOT prime; $\equiv 1 \pmod 4$ but composite — FAIL/SKIP.
- Kronecker: $13964 = 4 \cdot 3491$; $3491 \equiv 3 \pmod 4$ NOT Hadamard-admissible; 3491 (odd prime) — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_13964 Paley-I	Paley-I q=13963	0	2491.6549

Object	Method	max|HHˆT-nI|	Wall (s)
H_13964 Paley-II	FAIL/SKIP (q=6981=3*13*179 NOT prime; ===1 mod 4 but composite)	–	–
H_13964 Kronecker	FAIL/SKIP (4*3491; 3491===3 mod 4 NOT Hadamard-admissible; 3491 (odd prime))	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
12488 in SPK	**True** (Paley-I q=13963)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false

- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3^271317 *NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59^2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2^2*59^2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | *q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2^2*3*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | next Paley-I DIRECT: q=13967->n=13968 (prime ===3 mod 4; first prime ===3 mod 4 after 13963; also first n===0 mod 4 after 13964 with any classical Paley route; Paley-II q=6983 prime but ===3 mod 4 -> FAIL/SKIP (Paley-II needs q===1 mod 4); Kronecker 13968=4*3492; 3492=2^2*3^2*97 ===0 mod 4 Hadamard-admissible (H_3492 via Paley-I q=3491 prime ===3 mod 4; also 2*6984 with H_6984 via Paley-I q=6983) -> Kronecker ELIGIBLE note-only (Paley-I is claim path; do NOT double-construct); 13968=2?*3^2*97 ===0 mod 16 – NOT the 13964/13932 odd-m shape) (or gap 2052 / Track B)